Technology Stack

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React 18 + Vite, CSS	Component-based UI, very fast dev server and build, simple deployment on Vercel.
2	Backend / Server-Side	Python 3 + FastAPI (Uvicorn)	Fast to build, automatic API docs at /docs, Pydantic validation, easy integration with AI libraries.
3	Database / Data Storage	Supabase (PostgreSQL)	Managed relational database with JSONB columns for itineraries and a free tier.
4	Cloud / Hosting / Deployment	Render (backend), Vercel (frontend)	Free tiers, deploy directly from GitHub, automatic redeploy on every push.
5	Version Control & CI/CD	Git + GitHub	Source control, and the push-to-deploy integration used by Render and Vercel.
6	Third-Party APIs / Other Tools	Groq API (Llama / gpt-oss models), python-dotenv, supabase-py	Groq gives very low-latency LLM inference with JSON mode, which generates the itinerary; dotenv and env variables keep secrets out of code.